Original Article

Developing Biosorbents from Agricultural Waste: Fava Bean Shells for Efficient Dye Removal

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ABSTRACT

In this study, two biosorbents were developed from fava bean shells (FBS): dried fava bean shells (DFBS) and fava bean shell charcoal (FBSC) and their adsorption performance for Congo red (CR) dye was compared. Adsorption was evaluated under varying conditions of temperature, initial dye concentration (C_0), adsorbent dose, contact time, and pH. At pH < 4, FBSC exhibited superior adsorption capacity (Q_e) and removal efficiency (99%) compared to DFBS (90%), highlighting the enhancement achieved through carbonization. Adsorption data were fitted to seven isotherm models; although the Langmuir model, showed high correlation, the calculated monolayer capacities differed from experimental maxima due to heterogeneous surface properties and broad concentration ranges, while Freundlich and Halsey models provided a physically meaningful description ($R^2 > 0.9$). Kinetic studies indicated that the pseudo-second-order model best described the adsorption process ($R^2 = 0.9928$ for DFBS, 0.9998 for FBSC), suggesting that chemisorption is dominant, with reasonable fits also obtained using the Elovich and intra-particle diffusion models ($R^2 > 0.9$). Thermodynamic analysis, based on dimensionless equilibrium constants, revealed that adsorption onto FBSC is spontaneous and endothermic with increased randomness at the adsorbent–adsorbate interface, whereas adsorption onto DFBS is spontaneous but slightly exothermic, reflecting moderate interactions. These findings indicate that FBSC is an efficient, economical, and environmentally friendly biosorbent for dye removal from aqueous solutions.

Introduction

Textile dyes are extensively used in industries such as printing, cosmetics, and textiles, and the discharge of dye-polluted wastewater into aquatic systems poses serious environmental and public health concerns [1,2]. Many synthetic dyes, particularly azo dyes such as Congo red (CR), are highly stable and resistant to biodegradation,

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oxidation, light, and heat, leading to their long-term persistence in water bodies. Exposure to these compounds has been associated with numerous adverse health effects including allergic reactions, skin irritation, genetic mutations, reproductive toxicity, and systemic absorption problems [3]. CR, in particular, is a well-known toxic and mutagenic dye that decomposes into benzidine, a recognized human carcinogen. This degradation product can cause kidney failure, respiratory allergies, corneal damage, and severe ecological harm to aquatic organisms [4–6]. Consequently, CR has been banned in several countries despite its widespread industrial use.

A variety of physicochemical and biological methods such as precipitation, ozonation, coagulation/flocculation, reverse osmosis, oxidation, ion exchange, irradiation, and membrane separation have been explored for the removal of dyes from wastewater [7–11]. However, these conventional approaches often suffer from limitations related to cost, energy consumption, and operational complexity, particularly for high-strength dye effluents. In contrast, adsorption has emerged as one of the most efficient and economical techniques due to its simplicity, high removal efficiency, and reusability potential [12]. The success of adsorption processes strongly depends on the physicochemical characteristics of the adsorbent material.

A wide range of natural and agricultural by-products has been studied as low-cost adsorbents, including orange peels, rice husks, sugarcane bagasse, peanut hulls, olive leaves, sawdust, loofah sponges, etc. [13–19]. Recent research has also explored biosorbents and green-synthesized nanomaterials as sustainable solutions for dye and metal removal from wastewater. For instance, Albo Hay Allah and Alshamsi [20] fabricated zinc oxide nanoparticles using an eco-friendly plant-based route and demonstrated their high efficiency in adsorbing both anionic and cationic dyes. In a related study, Albo Hay Allah and Alshamsi [21] developed ZnO–activated carbon nanocomposites through a green synthesis approach, achieving remarkable adsorption capacities for CR and Malachite Green. Hussain *et al.* [22] prepared sulfur- and nitrogen-doped carbon nanospheres from Hibiscus sabdariffa extract and reported their effective performance in Pb^{2+} removal. Imessaoudene *et al.* investigated natural zeolite for CR adsorption and highlighted the strong influence of pH and operating conditions on removal efficiency [23]. Urooj *et al.* utilized Phyllanthus emblica fruit powder as a low-cost adsorbent for crystal violet and confirmed its heterogeneous surface and favorable adsorption behavior [24].

These studies collectively demonstrate the growing interest in low-cost, environmentally friendly materials for efficient pollutant removal, supporting the continued development of biosorbents derived from agricultural waste. Despite these advancements, the use of fava bean shells (FBS)—an abundant agri-food by-product particularly common in Mediterranean and North African countries—remains largely unexplored as a biosorbent precursor. The natural composition of FBS, which is rich in cellulose, hemicellulose, lignin, and phenolic groups, provides a diverse array of functional sites capable of interacting with dye molecules through electrostatic attraction, hydrogen bonding, and π–π interactions. These intrinsic characteristics make the raw material itself an important contributor to adsorption behavior and differentiate it from other biomass sources traditionally used in dye removal studies. Furthermore, transforming discarded FBS into functional adsorbents adds value to agricultural residues and aligns with circular economy principles by converting food processing waste into sustainable treatment materials [3,8].

The research gap addressed in this study lies in the absence of systematic comparative data on raw and carbonized FBS and how their natural composition and thermal transformation influence adsorption efficiency. While numerous biosorbents have been reported in the literature, limited information exists on the extent to which carbonization modifies the surface chemistry, pore structure, and thermodynamic properties of FBS-derived materials. This work therefore provides a direct comparison between unmodified (DFBS) and carbonized FBS to clarify the role of the natural source and thermal treatment in governing adsorption behavior toward CR dye.

Therefore, the objective of this study is to develop and characterize two biosorbents derived from FBS (DFBS and FBSC) and to evaluate their efficiency in removing CR dye from aqueous solution. By analyzing adsorption behavior under different operational parameters (pH, contact time, temperature, dye concentration, and adsorbent dosage) and applying multiple isotherm and kinetic models, this study provides new insights into the valorization of an underutilized agricultural by-product. The findings are expected to inform the sustainable design of cost-effective, eco-friendly biosorbents for industrial wastewater treatment.

Materials and Methods

Biosorbent Material

After being purchased from shops in the Misurata area, the FBS were thoroughly cleansed with tap water to ensure they were free of any foreign objects and water-soluble contaminants. The biomass was cleaned, crushed into a fine powder, allowed to dry in the sun for two weeks, and then washed several times with distilled water. Following processing, the material was dried for 48 hours at 60 °C in an oven. A sieve that met the standards of the American society for testing and materials (ASTM) was used to create particles between 200 and 250 µm in size from the dried and powdered biomass. This adsorbent material was designated as DFBS. 50 g of the powdered adsorbent, carefully measured, were placed in crucibles and carbonized for an hour at 550 °C. The material was carefully cleaned and dried after carbonization. The designation for this carbonized adsorbent was FBSC.

Adsorption Procedure

Batch sorption experiments were used to adjust several sorption parameters in order to achieve maximal adsorption, including solution contact duration (0–60 minutes), pH (2–11), adsorbent dose (0.1–3.5 g), starting dye concentration (1–300 mg/L), and temperature (25–55 °C). While the other parameters remained constant, each one was varied separately. The pH of each solution was adjusted using a Jenway 3510 pH meter, and absorbance was measured with a Jenway 6305 UV–Vis spectrophotometer at 490 nm. Each experiment was conducted using 50 mL of dye solution, which was agitated at 175 rpm. After reaching equilibrium, the supernatant solution was filtered using Whatman filter paper No. 1, and the final absorbance was recorded to determine the remaining dye concentration. The C_0 was measured at the same wavelength before the adsorbent was added, and the concentrations before and after adsorption were determined using a previously developed calibration curve. All adsorption experiments were conducted in triplicate under identical conditions, and the reported values represent the mean ± standard deviation to ensure reproducibility and statistical reliability of the results.

Evaluation of Adsorption Efficiency

To evaluate the efficiency of the adsorption process, the equilibrium dye Q_e and the percentage removal of dye (%R) onto DFBS and FBSC were computed using the following formulas [25].

$$\%R = (C_0 - C_e)/ C_0 \times 100 \quad (1)$$

$$Q_e = ((C_0 - C_e) \times V)/M \quad (2)$$

Where, m is the biomass mass (g), V is the solution volume (L), and C_0 and C_e stand for the initial and equilibrium dye concentrations (mg/L).

Adsorption Equilibrium Study

To ascertain the relationship between the concentration of adsorbate in the equilibrium solution and the quantity of adsorbate retained on the adsorbent at a constant temperature, a variety of isotherm models were employed to examine the experimental results. Langmuir, Freundlich, Temkin, Dubinin-Radushkevich, Elovich, Jovanovic, and Halsey were the isotherm models used [26].

The linear form of the Q_m, which theoretically explains the development of a monolayer of adsorbate on the adsorbent surface with no further adsorption beyond this layer, is as follows [27].

$$\frac{1}{Q_e} = \frac{1}{Q_m} + \frac{1}{bQ_e\,Q_m} \quad (3)$$

Where, b is the Langmuir isotherm constant (L/mg) and Q_m is the maximal monolayer Q_e (mg/g).
The following is the linear form of the Freundlich equation that accounts for surface heterogeneity and the exponential distribution of active sites and their energy [4]:

$$logQ_e = logK_f + \frac{1}{n} logC_e \quad (4)$$

The heterogeneity component in this model is denoted by 1/n, while the Freundlich constant is represented by K_f (L/g).
The following is the expression for the linear version of the Temkin model, which postulates that interactions between the adsorbent and adsorbate cause the heat of adsorption to drop linearly with surface coverage [13].

$$Q_e = BTlnK_T + BTlnC_e \quad (5)$$

Where, B_T is the Temkin isotherm constant related to the heat of adsorption, and K_T is the Temkin isotherm equilibrium binding constant (L/g).
The adsorption mechanism based on a Gaussian energy distribution across a heterogeneous surface is described by the Dubinin-Radushkevich (D-R) model, which provides the following linear form:

$$\ln Q_e = \ln Q_m - \beta\,\varepsilon^2 \quad (6)$$

Where, ε (J/mol) stands for the Polanyi potential, and β (mol^2/J^2) is the D-R constant.
The following formula can be used to get the Polanyi potential (ε):

$$\varepsilon = RTLog\left(1 + \frac{1}{C_e}\right) \quad (7)$$

Where, T (K) is the absolute temperature and R (J/mol.K) is the gas constant.
The linear form of the Jovanovic model, which assumes that adsorbate interactions lead to a reduction in Q_e as surface coverage rises [28], appears to be as:

$$lnQ_e = lnQ_m + KJC_e \quad (8)$$

The Jovanovic isotherm constant (L/g) is denoted by K_J in this case.
The chemisorption process on heterogeneous surfaces is commonly described using the Elovich model, which accounts for variations in adsorption energy during the process [18]. Its linear version may be found in the following equation:

$$lnC_e/Q_e = lnK_E\,Q_m - Q_m/Q_e \quad (9)$$

Where, K_E (L/mg) is the Elovich constant
Adsorption in a multilayer system is evaluated using the Halsey isotherm [28]. The following equation represents the model:

$$lnQ_e = \frac{1}{n_H} lnK_H - 1/n_H lnC_e \quad (10)$$

Where, K_H (L/mg) and n_H are the constants of the Halsey model.

Adsorption Kinetic Study

The sorption kinetics of CR onto DFBS and FBSC were evaluated by examining the adsorption rate, mass transfer mechanisms, and time required to reach saturation for the adsorbent. The experimental data were fitted to a variety of kinetic models, including Elovich, pseudo-second-order, intra-particle diffusion, and pseudo-first-order models [13].
The pseudo-first-order kinetic model is represented by the following equation:

$$ln(Q_e - Q_t) = ln(Q_e) - \frac{k_1}{2.303} \times t \quad (11)$$

Where, Q_t (mg/g) is the Q_e at time t and k_1 (1/min) is the pseudo-first-order model's rate constant.
The pseudo-second-order rate equation, commonly employed in the analysis of adsorption kinetics, is as follows [18]:

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{t}{Q_e} \quad (12)$$

Where, k_2 (g/mg.min) is the pseudo-second-order rate constant for adsorption and Q_e (mg/g) is the quantity of solute adsorbed at equilibrium.

The intra-particle diffusion model describes the rate-limiting stage of adsorption, which is the diffusion of adsorbate molecules through the pores of the adsorbent. The intra-particle diffusion model equation is commonly expressed as follows [26]:

$$Q_t = k_d t^{1/2} + C \quad (13)$$

C (mg/g) is associated with the boundary layer effect, which quantifies the thickness of the boundary layer around the adsorbent particles.

The adsorption of dissolved chemicals from liquid solutions has been well described by the Elovich equation, despite the fact that it was first developed to study the kinetics of chemical adsorption of gases onto solids [25]. The linear version of the Elovich equation is as follows:

$$Q_t = \frac{1}{\beta}\ln(\alpha\beta) + \frac{1}{\beta}\ln t \quad (14)$$

Where, β (g/mg) is associated with surface coverage and the chemisorption activation energy, and α (mg/g.min) is the initial rate of sorption.

Adsorption Thermodynamic Study

Thermodynamic analysis is necessary to ascertain whether a biosorption process occurs spontaneously. The change in Gibbs free energy, or ΔG^o, is a crucial component in assessing spontaneity and may be calculated using the following formulae [29,30]:

$$\Delta G = \Delta H - T\Delta S \quad (15)$$

ΔH^o and ΔS^o are calculated using the Van't Hoff equation as follows [18]:

$$\ln K_d = \frac{\Delta S}{R} - \frac{\Delta H}{RT} \quad (16)$$

Furthermore, the following formula [13] is used to obtain the distribution coefficient (K_d):

$$K_d = \frac{Q_e}{RC_e} \quad (17)$$

The values for entropy change (ΔS°) and enthalpy change (ΔH°) were calculated from the slope and intercept of the plot of $\ln K_d$ versus 1/T. The ΔG° shift for sorption was then calculated using Equation 15.

Results and Discussion

Investigation of Optimal Conditions for Dye Adsorption

The main factors affecting adsorption efficiency were investigated in the present research, including contact time (t), solution pH, adsorbent dose (m), and C_o. The %R and Q_e were determined to assess adsorption performance. Figure 1(a-d) shows how these factors affect the process.

The impact of contact time on removal efficiency is demonstrated in Figure 1a, which also illustrates how rapidly the adsorption process achieved equilibrium for both DFBS and FBSC. The C_0 and adsorbent dosage were maintained at 50 mg/L and 0.5 g, respectively, in both cases. Since the initial removal rate for FBSC was 70%, which was much greater than the 12% found for DFBS, adsorption occurred more rapidly with FBSC than with DFBS. While DFBS needed 30 min to reach equilibrium with a lower removal efficiency of 52%, FBSC reached equilibrium with an 87% removal efficiency in 20 min. These findings highlight the greater efficacy of fava bean shell-derived charcoal (FBSC) in eliminating CR dye.

The effect of the adsorbent dose is an essential indicator to assess adsorption experiments. Since Q_e directly represents the number of active adsorption sites on the adsorbent surface, it is more informative to determine it in such investigations than the removal percentage (%R). Figure 1b shows how the dose of the adsorbent affects the Q_e for DFBS and FBSC. The results show that as the amount of adsorbent increases, the Q_e decreases. The dye concentration and solution volume for this experiment were kept at 50 mg/L and 50 mL, respectively. The Q_e approached 20 mg/g when 0.1 g of adsorbent was employed; however, when the adsorbent quantity was raised to 3.5 g,

it significantly decreased to 0.5 mg/g. These patterns held true for surfaces fabricated from FBSC and DFBS. Although both materials include active sites for adsorption, the observed decrease in Q_e with greater adsorbent doses implies that these sites become less accessible as the amount of adsorbent rises. Particle agglomeration is the cause of this process, which limits the interaction between the dye molecules and the adsorbent and causes surface coverage [31].

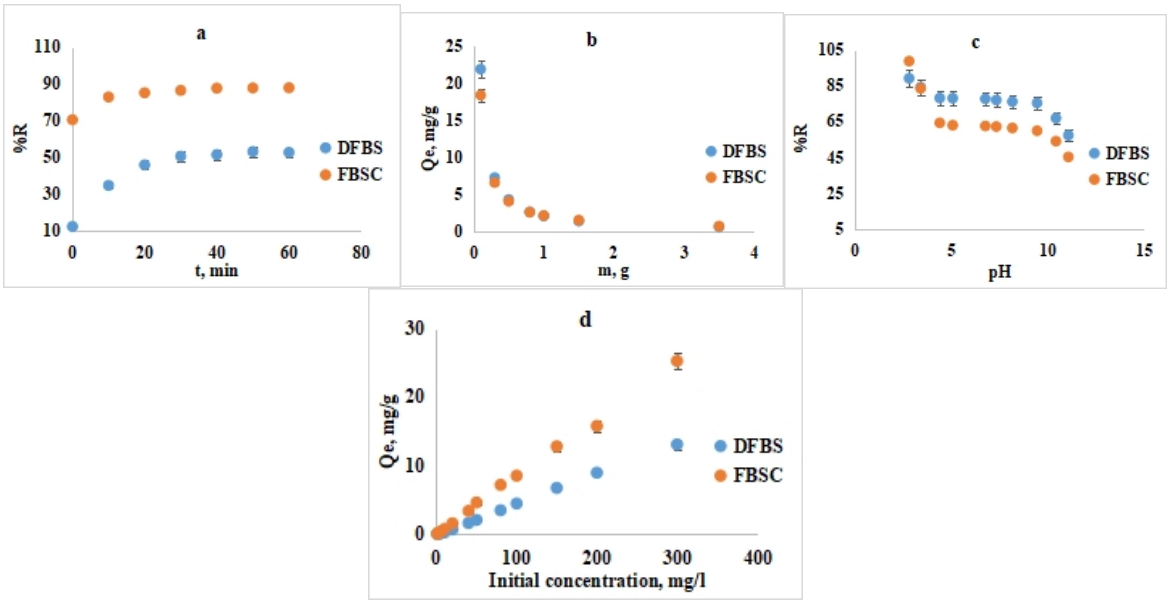


Figure 1: Parameters affecting adsorption of CR onto DFBS and FBSC: (a) time, (b) dose, (c) pH, and (d) concentration

The pH of the solution is a crucial factor in adsorption operations as it significantly affects the surface charge of the adsorbent. In extremely basic conditions, the adsorbent's surface becomes negatively charged, while in extremely acidic conditions, it becomes positively charged. This charge shift has a direct effect on the interaction between the adsorbent and the target adsorbate, underscoring the importance of investigating pH effects in adsorption studies [4,13]. As observed in Figure 1c, the dye removal effectiveness (%R) at various pH values was examined to investigate the effect of pH on the adsorption of CR dye onto DFBS and FBSC surfaces. Under acidic conditions (pH < 4), the removal effectiveness increases significantly, reaching 90% for DFBS and 99% for FBSC, according to the results. On the other hand, the removal efficiency decreased to 60% for FBSC and 75% for DFBS at basic conditions (pH > 9). These results are consistent with CR's characteristics as an anionic dye. The adsorbent surface becomes positively charged at pH values lower than 4, which promotes a strong electrostatic interaction between the surface and the anionic dye molecules and improves adsorption. Conversely, the surface becomes negatively charged at pH values higher than 9, resulting in electrostatic repulsion and, ultimately, a reduction in adsorption effectiveness. Furthermore, under different pH levels, the adsorption of organic molecules may be influenced by additional intermolecular interactions such hydrogen bonds and van der Waals forces [32].

The effect of the C_0 on the adsorption performance of CR dye onto DFBS and FBSC surfaces was investigated using 1.0 g of adsorbent. The correlation between Q_e and C_0 is depicted in Figure 1d. According to the results, the adsorption capacities of both adsorbents increased as the initial dye concentration increased from 1.0 to 300 mg/L.. The highest adsorption capabilities that were attained were 26 mg/g for FBSC and 14 mg/g for DFBS. According to this pattern, there are still enough

adsorption sites on the surfaces at higher concentrations, and the concentration of dye molecules increases the force that drives their interaction with the active sites. This driving force is less at lower concentrations, resulting in a decreased Q_e [13,16]. Determining the adsorption isotherm and comprehending the adsorption mechanism requires analyzing the impact of C_0 on Q_e. This information offers important insights into how the adsorbent behaves at different concentrations and its possible applications in real-world scenarios [28,31].

The results clearly indicate that carbon derived from FBSC demonstrated a higher adsorption efficiency for CR dye than DFBS, even under the studied optimal conditions. Remarkably, this improved performance was observed without any activation treatment being applied to the FBSC.

Investigation of Adsorption Isotherms

The experimental data obtained for CR dye adsorption on DFBS and FBSC at constant temperature were compared with several isotherm models [28]. The corresponding plots are presented in Figure 2.

The correlation coefficient (R^2) was used to assess each isotherm model's suitability for the experimental data. The findings showed that the isotherm models were in the following order of suitability: Halsey, Freundlich, and Langmuir. The R^2 for the Temkin, D-R, Elovich, and Jovanovic models were all less than 0.9, indicating that none of them were suitable for explaining the experimental data. The Elovich model produced the least favorable match.

The monolayer adsorption process with uniform adsorption sites was proposed by the Q_m, which showed the highest R^2 values. There was minimal interaction between the dye molecules and the adsorbent surface, most likely because of physical interactions like van der Waals forces, as shown by the Langmuir adsorption constant, which measures the affinity between the adsorbent and the adsorbate, being less than 1. The separation factor (R_L), which provides information about the adsorption process, is a crucial metric for the Q_m. For both DFBS and FBSC, the R_L values ranged from 0 to 1 (0.039 to 0.042), suggesting that the adsorption process operates well [27,28].

The presence of heterogeneous adsorption sites was suggested by the Freundlich model, which also produced superb fits with R^2 values higher than 0.9. While the value of the parameter n, which represents the heterogeneity and intensity of adsorption, was near 1, suggesting a linear adsorption process, the K_f was quite high, indicating large Q_e on the surfaces [28].

Multilayer adsorption was suggested by the Halsey model, which also showed good correlation values ($R^2 > 0.9$). Strong interactions between the adsorbent and the adsorbate were suggested by the Halsey constant (K_H), which was much higher than 1, supporting the existence of cooperative adsorption. Interestingly, the experimental results and the Q_e values predicted by the Halsey model agreed well [33]. Table 1 lists the constant values for each model together with the related R^2 values.

The higher discrepancy between the maximum Q_e determined experimentally and that calculated from the Q_m, despite the high R^2 values, can be attributed to the assumptions and limitations of the Q_m. Langmuir assumes monolayer adsorption on uniform sites with minimal interaction between adsorbed molecules, whereas the dry fava bean shells (DFBS) and FBSC adsorbents possess heterogeneous surfaces with multiple functional groups. Additionally, the adsorption experiments were conducted over a wide concentration range (1–300 mg/L), which can further distort the linearized Langmuir fit and lead to underestimation of Q_m relative to the observed $Q_{e,max}$. In contrast, the Freundlich and Halsey models, which account for heterogeneous and multilayer adsorption, provide a more physically meaningful description of the adsorption behavior. This explains why the experimental $Q_{e,max}$ values (14 mg/g for DFBS and 26 mg/g for FBSC) are substantially higher than the Langmuir Q_m derived from linear fitting [13,27].

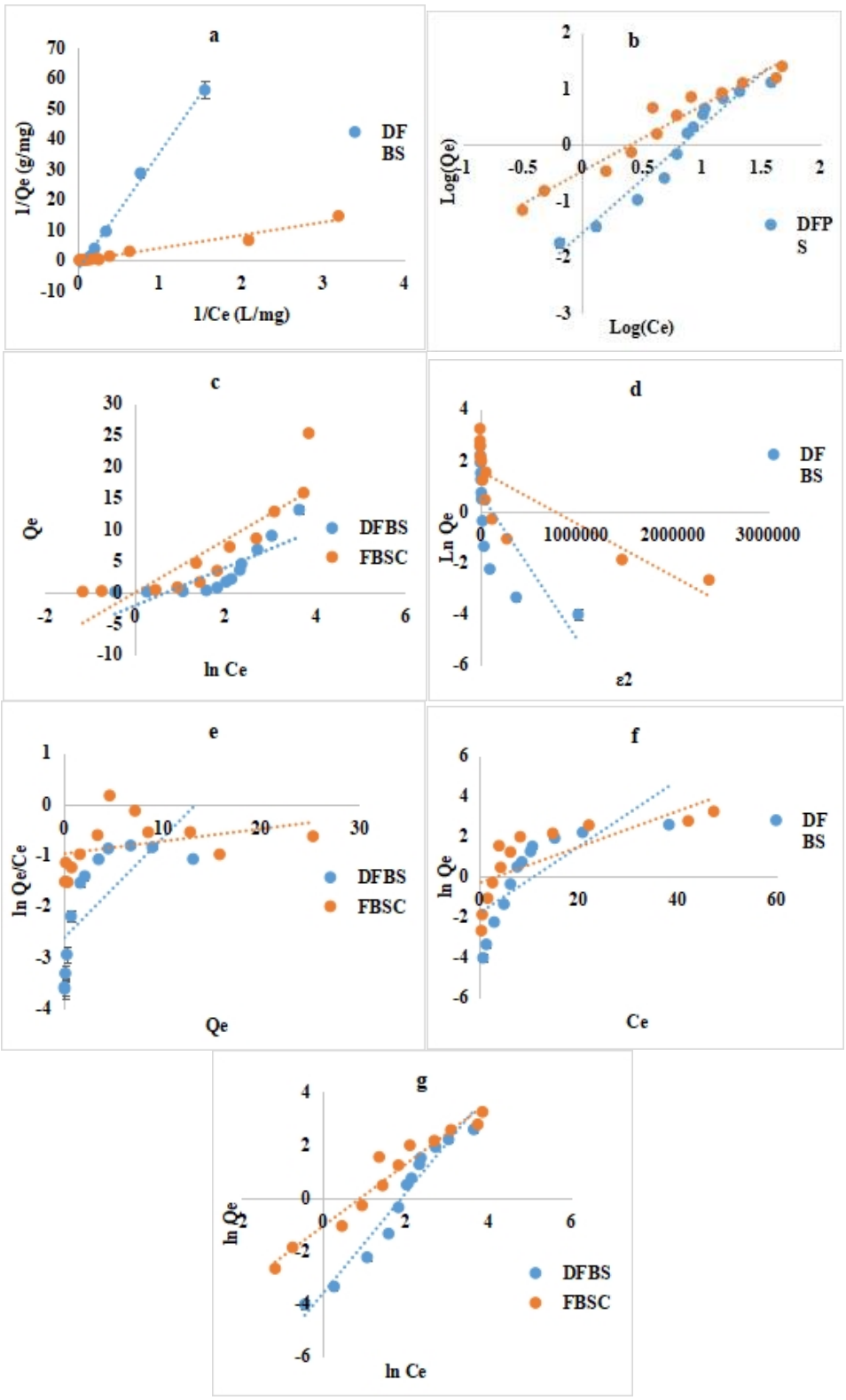


Figure 2: Plots of various adsorption isotherm models for CR adsorption onto DFBS and FBSC: (a) Langmuir, (b) Freundlich, (c) Temkin, (d) Dubinin-Radushkevich (D-R), (e) Elovich, (f) Jovanovic, and (g) Halsey

Table 1: Parameter values for various isotherms models for adsorption of CR

Langmuir	**Cal. Q_m mg/g**	**Obs. Q_e max mg/g**	**b L/mg**	**R_L**	**R^2**
DFBS	0.321	3.46	0.081	0.039	0.9939
FBSC	3.049	6.70	0.077	0.042	0.9681
Freundlich	**K_f $(mg/g)(L/mg)^{1/n}$**		**n g/L**		**R^2**
DFBS	36.87		0.53		0.9504
FBSC	2.94		0.86		0.9456
Temkin	**B_T J/mol**		**K_T L/mg**		**R^2**
DFBS	3.00		2.07		0.6751
FBSC	14.17		1.03		0.7329
D-R	**Q_m mg/g**		**β mol^2/J^2**		**R^2**
DFBS	2.02		6×10^{-6}		0.5733
FBSC	4.87		2×10^{-6}		0.6512
Elovich	**Q_m mg/g**		**K_E L/mg**		**R^2**
DFBS	1.67		5.12		0.5238
FBSC	1.02		40.98		0.1324
Jovanovic	**Q_m mg/g**		**K_J L/mg**		**R^2**
DFBS	6.03		0.164		0.5963
FBSC	1.36		0.089		0.5591
Halsey	**n_H**		**K_H L/mg**		**R^2**
DFBS	0.53		6.76		0.9504
FBSC	0.86		2.52		0.9456

Investigation of Adsorption Kinetics

With R^2 of 0.9928 and 0.9998, respectively, the pseudo-second-order model displayed the best match to the experimental data in the kinetic examination of CR dye adsorption onto DFBS and FBSC surfaces (Figure 3). The model's predicted Q_e values, which were determined to be 1.2 and 4.4 mg/g for DFBS and FBSC, respectively, closely matched the actual values, which were 1.5 and 4.5 mg/g. The good fit of the pseudo-second-order model is consistent with a chemisorption contribution; however, kinetic fitting alone is insufficient to establish the adsorption mechanism [34].

High R^2 (0.9841 and 0.9757) were also displayed by the Elovich model, indicating that it was suitable for describing the adsorption kinetics. The comparatively high β values suggest slower adsorption rates, which might be caused by a chemical adsorption process, restricted diffusion, or a lack of accessible active sites. On the other hand, the moderate to low α values indicated that the initial adsorption rates were poor and that an activation step could be necessary to improve the process.

The relatively small but acceptable R^2 for the Intra-Particle Diffusion model showed that, while not the only rate-limiting step, intra-particle diffusion played a role in the adsorption mechanism. The comparatively low intra-particle diffusion rate constant (K_d) results indicated slow intra-particle diffusion, which might be caused by a low driving force or a small pore size. The non-zero intercept indicates a contribution from boundary-layer resistance, while the linearity of the intra-particle diffusion plot should be considered when assessing whether diffusion is the sole rate-limiting step. The FBSC's comparatively high C value, on the other hand, indicated a stronger boundary layer effect where surface adsorption predominated over intra-particle diffusion [34,35].

The poor R^2 of the pseudo-first-order model, on the other hand, highlighted unsuitability of the model to describe the adsorption kinetics for both adsorbents. This finding adds credibility to the idea that the pseudo-second-order model

and chemisorption processes provide a more realistic explanation of the adsorption mechanism. Table 2 presents all parameter values for each model along with the related R^2 values.

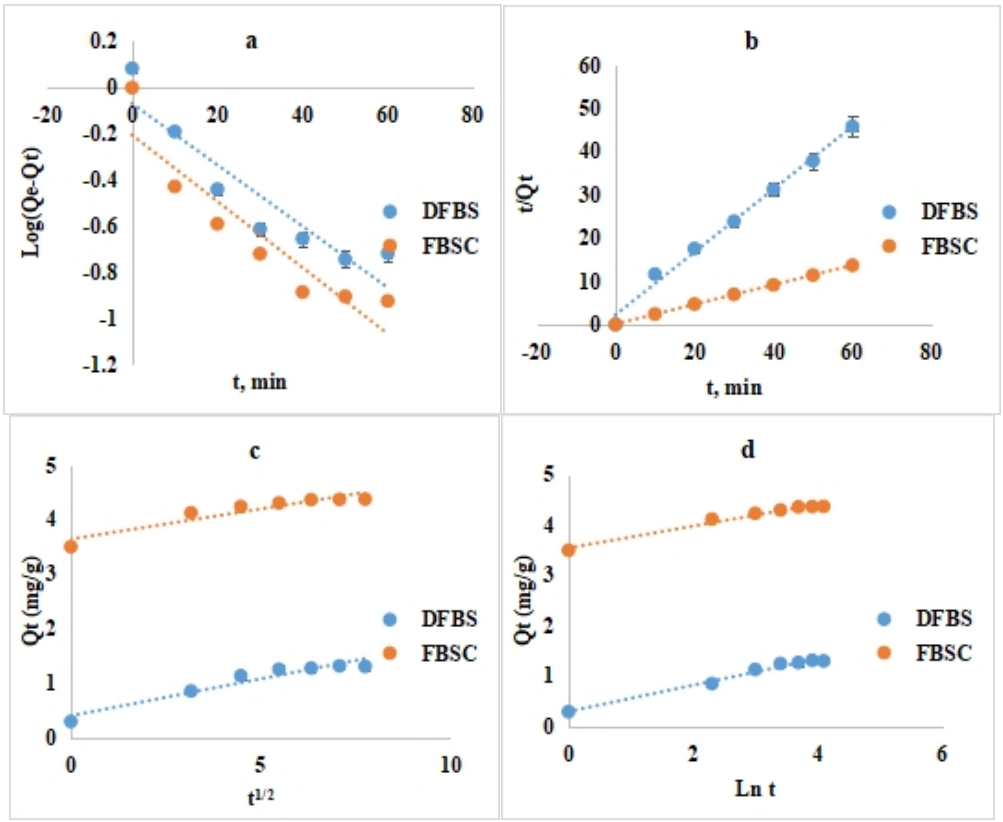


Figure 3: Plots of various adsorption kinetic models for CR adsorption onto DFBS and FBSC: (a) pseudo-first-order, (b) pseudo-second-order, (c) intra-particle diffusion, and (d) Elovich

Table 2: Parameter values for various kinetic models for the adsorption of CR

Pseudo-First-Order				
Adsorbent	**k_1 1/min**	**Q_e mg/g**		**R^2**
		Cal.	**Exp.**	
DFBS	0.0132	1.179	1.50	0.8601
FBSC	0.0143	1.613	4.50	0.8564
Pseudo-Second-Order				
Adsorbent	**k_2 g/mg.min**	**Q_e mg/g**		**R^2**
		Cal.	Exp.	
DFBS	0.2418	1.374	1.50	0.9928
FBSC	0.500	4.407	4.50	0.9998
Elovich				
Adsorbent	**β g/mg**	**α mg/g.min**		**R^2**
DFBS	3.818	0.830		0.9841
FBSC	4.583	1.111		0.9757

Adsorbent	Intra-Particle Diffusion		
	K_d mg/g.min$^{1/2}$	C mg/g	R^2
DFBS	0.1356	0.4015	0.9245
FBSC	0.1109	3.645	0.8833

Investigation of Adsorption Thermodynamics

To understand further about the nature of the adsorption process, the thermodynamic parameters for the adsorption of CR dye onto DFBS and FBSC were established. Key details regarding the spontaneity, heat exchange, and molecular interactions involved in the adsorption mechanism are revealed by the results, which include ΔH°, ΔS°, and ΔG° [34]. The thermodynamic parameters are listed in Table 3. The corresponding Van't Hoff plots are presented in Figure 4. The thermodynamic analysis of CR dye adsorption onto DFBS revealed that the process is slightly exothermic, as indicated by the negative enthalpy change (ΔH° = −3.49 kJ/mol). This suggests that adsorption is primarily governed by weak physical interactions, such as van der Waals forces. The small positive entropy change (ΔS° = 0.034 kJ/mol.K) implies a slight increase in randomness at the adsorbent–adsorbate interface, likely due to minor reorganization of dye molecules during adsorption. The negative Gibbs free energy values (ΔG° = −13.63 to −14.31 kJ/mol over 298–318 K) indicate that the adsorption is spontaneous under the studied conditions, though the relatively low magnitude of ΔG° reflects moderate affinity between the dye molecules and the DFBS surface. These results suggest that, while DFBS can adsorb CR dye, its adsorption efficiency may be enhanced by surface modification or activation to create more active sites and strengthen interactions [34–36].

In contrast, the thermodynamic parameters for CR dye adsorption onto FBSC indicate a strongly endothermic and spontaneous process (ΔH° = 48.61 kJ/mol; ΔG° = −17.63 to −22.08 kJ/mol; ΔS° = 0.222 kJ/mol.K). The positive ΔH° reflects an energy-driven process, potentially involving chemisorption through covalent interactions or hydrogen bonding. The substantial positive ΔS° suggests increased randomness at the solid–liquid interface, likely due to displacement of water molecules and reorganization of dye molecules during adsorption. The increasingly negative ΔG° values with rising temperature confirm the enhanced spontaneity at higher temperatures, consistent with strong adsorbent–dye interactions. Overall, these results highlight FBSC as an efficient and promising adsorbent for practical removal of wastewater dye, even without additional surface activation [34–37].

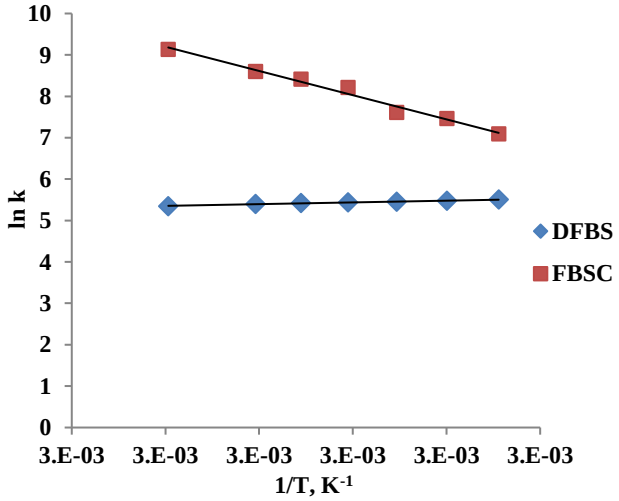


Figure 4: Van't Hoff plots for CR adsorption onto DFBS and FBSC

Table 3: Thermodynamic parameters for the adsorption of CR

Dye	Temperature (K)	ΔG° (kJ/mol)	ΔH° (kJ/mol)	ΔS° (kJ/mol. K)	R^2
DFBS	298	-13.63	-3.49	0.034	0.9897
	303	-13.80			
	308	-13.97			
	313	-14.14			
	318	-14.31			
FBSC	298	-17.63	48.61	0.222	0.9831
	303	-18.74			
	308	-19.85			
	313	-20.97			
	318	-22.08			

The calculated thermodynamic parameters (Table 3) confirm that CR dye adsorption onto both adsorbents is spontaneous under the studied conditions, with FBSC exhibiting endothermic and highly favorable adsorption, while DFBS shows slightly exothermic and moderate adsorption. The positive ΔS° values, particularly for FBSC, reflect increased randomness at the interface and correlate with strong dye–surface interactions. To contextualize these findings, Table 4 compares the maximum adsorption capacities (Q_{max}) of DFBS and FBSC with other biosorbents reported in the literature, highlighting the superior adsorption performance of FBSC.

Comparison of Adsorption Performance with Reported Biosorbents

To better assess the performance of the developed biosorbents, the Q_{max} obtained in this study were compared with those reported for other low-cost agricultural and natural adsorbents used for CR removal (Table 4). The FBSC biosorbent (26.0 mg/g) outperformed a range of commonly studied materials, including Vangueria infausta fruit pericarp (7.91 mg/g), activated carbon–Juncus effusus composites (7.00 mg/g) [38], and Azadirachta indica leaves (9.98 mg/g) [2]. It also exceeded the adsorption capacities reported for most activated glauconite materials (5.1–13.2 mg/g) [39]. Although certain highly engineered materials—such as modified orange-peel biochar (290.1 mg/g) and unmodified orange biochar (155.2 mg/g) [40] demonstrated very high adsorption capacities, these materials undergo chemical modification or activation that places them in a different category than simple agricultural-waste biosorbents. In comparison, FBSC still exhibited competitive performance considering its low-cost preparation and absence of chemical activation.

Zeolite has also shown exceptionally high capacity (232 mg/g) under strongly acidic conditions (pH 2–3) [23]; however, this mineral-based adsorbent differs significantly from biomass-derived materials. Meanwhile, DFBS displayed a moderate Q_e of 14.0 mg/g, falling within the range of several reported agricultural biosorbents, such as potato peels (6.90 mg/g) and pea peels (16.40 mg/g) [41]. Overall, the comparison indicates that FBSC competes well with many biosorbents prepared from inexpensive agricultural residues and shows strong promise as an efficient, sustainable material for dye removal. Despite the promising adsorption performance of FBSC and DFBS, the present study has certain limitations. Structural, morphological, and textural characterizations were not conducted due to limited access to instrumentation, which constrains a detailed understanding of the adsorption mechanisms. Furthermore, experiments on adsorbent regeneration, reuse, and long-term stability were not performed. Future studies should focus on these aspects to evaluate the practical applicability and economic feasibility of fava bean shell-derived biosorbents. Additionally, exploring the performance of these materials in real wastewater and under continuous flow conditions could provide further insights into their suitability for industrial-scale dye removal applications.

Table 4: Comparison of adsorption capacities of selected natural biosorbents for CR dye

Biosorbent material	Q_{max} (mg/g)	pH	Temperature (°C)	[Ref.]
Azadirachta indica leaves	9.98	5	25	[2]
Zeolite	232.00	2-3	25	[23]
Vangueria infausta fruit pericarp	7.91	2	30	[10]
Glauconite (activated): GL.	13.2			
GLACT1M	13.1	7	25	[39]
GLACT2M	11.9			
GLACT3M GLACT4M	11.7			
	5.1			
Unmodified and modified orange-peel biochar: OBC	155.2	3	30	[40]
NOBC	290.1	3		
Activated Carbon-Juncus effusus Porous Fibril Composites	7.00	-	30	[38]
Solanum tuberosum (STP)	6.90	3	30	[41]
Pisum sativum peels (PSP)	16.40	2	30	
FBSC	26.0	4	25	The present work
DFBS	14.0	4	25	The present work

Conclusion

This study developed two biosorbents from FBS—DFBS and FBSC—for the removal of CR dye from aqueous solutions. Both materials exhibited efficient adsorption performance, reaching equilibrium within 20 minutes. Under optimal conditions (pH < 4), FBSC demonstrated a higher Q_e and removal efficiency (up to 99%) compared to DFBS (approximately 90%), highlighting the enhancement achieved through carbonization. Adsorption isotherm analysis revealed that, while the Q_m provided high correlation, the calculated monolayer capacities (Q_m) differed from the experimental maximum adsorption ($Q_{e,max}$) due to the heterogeneous nature of the adsorbent surfaces and the wide concentration range studied. Freundlich and Halsey models further confirmed the presence of heterogeneous and multilayer adsorption interactions, providing a more physically meaningful description of the adsorption process. Kinetic studies indicated that the pseudo-second-order model best described the adsorption behavior ($R^2 > 0.99$), suggesting that chemisorption may contribute substantially to the adsorption process. Thermodynamic analysis, performed using dimensionless equilibrium constants, showed that adsorption on FBSC is spontaneous and strongly endothermic, reflecting energy-driven adsorption with high affinity, while adsorption on DFBS is spontaneous but slightly exothermic, indicating moderate interactions. These findings are consistent with the experimental adsorption trends and highlight the distinct physicochemical characteristics of the two materials. This work demonstrates the feasibility of converting FBS into cost-effective, eco-friendly, and efficient adsorbents for dye-contaminated wastewater treatment. Future studies will focus on detailed material characterization, adsorption–desorption cycle testing, and optimization of surface properties to further elucidate adsorption mechanisms and enhance reusability, contributing to sustainable and circular waste management strategies.

Declaration of Competing Interest

The authors declared no competing financial or personal interests that could influence the findings presented in this study.

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